Question: A patch of lily pads doubles in size every day. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover exactly half of the lake?
[**Options**: A. 24 days, B. 47 days, C. 12 days, D. 36 days]

Chain of Thought

The lily pads cover the entire lake in 48 days. To find when they cover half the lake, I'll use half the time:
48 / 2 = 24 days
Therefore, the final answer is: A. 24 days

Tree of Thought

Branch 1: 100% coverage = 48 days
Branch 2: 50% coverage should be proportional
Branch 3: 50%×48=24 days
Therefore, the final answer is: A. 24 days

Our Method

Socratic Questioning:

1. What exactly does "doubles in size every day" mean mathematically?
2. Is the growth pattern linear or exponential?
3. If the patch covers the entire lake on day 48, what was its size on day 47?
4. How do we work backwards from exponential growth?

Answer:
By Question 1 & 2: "Doubles in size every day" describes exponential growth, not linear growth. The growth rate accelerates dramatically as time progresses.
By question 3 & 4: Since the patch doubles each day, going backwards means it halves each day:

- Day 48: 100% of the lake (full coverage)
- Day 47: 50% of the lake (half the size of day 48)

Mathematical verification: When we reverse exponential doubling, we get exponential halving:

$$\text{Day 47 size} = \frac{\text{Day 48 size}}{2} = \frac{100\%}{2} = 50\%$$

Therefore, the correct answer is: B. 47 days